Strategy-to-tool reference

High-Impact Teaching Math Tools Without the Price Tag

Use this reference

Start with the mathematical goal. Identify the strategy, a tool that serves it, and observable reasoning. Values below are the current MetaX weighted mean effect sizes; they are not tool ratings or promises of gains.

Teacher clarity | Digital Math Tools

Before comparing graphs, say: “Use the axis scale and the values to explain how the same counts can look different.”

Learning evidence: A learner states the goal and explains the success criteria.

MetaX influence: Teacher clarity; weighted mean d = 0.85. Verified 2026-10-27.

This synthesis concerns clarity across settings; naming a goal alone is not the full approach.

https://www.visiblelearningmetax.com/influences/view/teacher_clarity

Direct instruction | Digital Math Tools

Think aloud as you count three jumps from 4 on a number line. Ask learners to solve a similar example and explain the jumps.

Learning evidence: A learner explains each step in a similar new example.

MetaX influence: Direct instruction; weighted mean d = 0.56. Verified 2026-10-27.

This broad influence includes structured instructional programs. One demonstration is only a small illustration.

https://www.visiblelearningmetax.com/influences/view/direct_instruction

Argumentation | GraphSplat

Defend the zero-baseline graph using the insect counts, then discuss what a truncated baseline changes.

Learning evidence: A claim includes labels, values, and a response to another explanation.

MetaX influence: Argumentation / Classroom discussion; weighted mean d = 0.86. Verified 2026-10-27.

A short neighbor exchange models a discussion routine; it does not reproduce every condition in these studies.

https://www.visiblelearningmetax.com/influences/view/argumentation~classroom_discussion

Mathematical problem solving | GraphSplat

Plan equal-area garden samples, choose a display, and explain what the counts do and do not tell you.

Learning evidence: A learner chooses a representation and checks its fit to the question.

MetaX influence: Mathematics problem solving; weighted mean d = 0.79. Verified 2026-10-27.

The synthesis covers varied mathematical interventions and learners. Its estimate is not a prediction for one activity.

https://www.visiblelearningmetax.com/influences/view/mathematics_problem_solving

Corrective feedback | GraphSplat

If a learner says Garden B has twice as many insects, cue them to read the labels and calculate 15 minus 12.

Learning evidence: A revised explanation addresses the specific misconception.

MetaX influence: Feedback; weighted mean d = 0.5. Verified 2026-10-27.

The reported value is for the broader Feedback influence, not a separate estimate for corrective feedback or software feedback.

https://www.visiblelearningmetax.com/influences/view/feedback

Retrieval practice | Quiz and Flashcard Studio

Answer the three graph questions from memory, review the reasoning, and revisit a new example next week.

Learning evidence: A learner recalls and then corrects an explanation without looking back first.

MetaX influence: Practice testing; weighted mean d = 0.49. Verified 2026-10-27.

MetaX calls this Practice testing. A three-question conference activity illustrates the routine; it does not establish a learning gain.

https://www.visiblelearningmetax.com/influences/view/practice_testing

Transfer strategies | GraphSplat

Apply the same scale-checking routine to a campus temperature graph and explain what changes for a line graph.

Learning evidence: A learner adapts the reasoning to a new dataset or concept.

MetaX influence: Transfer strategies; weighted mean d = 0.89. Verified 2026-10-27.

Transfer depends on the prior learning and the distance between tasks. A reflection prompts planning, not proof of transfer.

https://www.visiblelearningmetax.com/influences/view/transfer_strategies

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